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Circulating cell-free RNA signatures for the characterization and diagnosis of myalgic encephalomyelitis/chronic fatigue syndrome

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People living with myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) experience heterogeneous and debilitating symptoms that lack sufficient biological explanation, compounded by the absence of accurate, noninvasive diagnostic tools. To address these challenges, we explored circulating cell-free RNA (cfRNA) as a blood-borne bioanalyte to monitor ME/CFS. cfRNA is released into the bloodstream during cellular turnover and reflects dynamic changes in gene expression, cellular signaling, and tissue-specific processes. We profiled cfRNA in plasma by RNA sequencing for 93 ME/CFS cases and 75 healthy sedentary controls, then applied machine learning to develop diagnostic models and advance our understanding of ME/CFS pathobiology. A generalized linear model with least absolute shrinkage selector operator regression trained on condition-specific signatures achieved a test-set AUC of 0.81 and an accuracy of 77%. Immune cfRNA deconvolution revealed differences in platelet-derived cfRNA between cases and controls, as well as elevated levels of plasmacytoid dendritic, monocyte, and T cell–derived cfRNA in ME/CFS. Biological network analysis further implicated immune dysfunction in ME/CFS, with signatures of cytokine signaling and T cell exhaustion. These findings demonstrate the utility of RNA liquid biopsy as a minimally invasive tool for unraveling the complex biology behind chronic illnesses.

cell-free RNA | diagnosis | ME/CFS | cell signaling | machine learning

Significance

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating illness that affects millions of individuals worldwide. Despite the growing prevalence of ME/CFS, patients suffer from the unavailability of laboratory diagnostic tests. Here, we establish cell-free RNA (cfRNA) as a minimally invasive bioanalyte for investigating circulating biomarkers and the pathobiology of ME/CFS. Using machine learning, we develop a cfRNA-based diagnostic model with high accuracy. We find evidence of immune system dysfunction in patients, with elevated levels of immune cell-derived transcripts as well as chronic inflammatory signaling pathways. These findings highlight the potential of circulating cfRNA to advance biomarker discovery and uncover disease mechanisms for ME/CFS.

Myalgic encephalomyelitis (ME), also referred to as chronic fatigue syndrome (CFS) and commonly referred to as ME/CFS, is a multisystem, heterogeneous, chronic illness. ME/CFS is characterized by persistent symptoms of postexertional malaise (PEM), extreme fatigue, unrefreshing sleep, cognitive impairment, and orthostatic intolerance (1, 2). An estimated 1.3% of the US population and 65 million individuals worldwide are affected by ME/CFS (3–5). Despite its high global prevalence, the underlying pathobiology of ME/CFS remains poorly understood. Currently, there are no clinically validated biomarkers for objective diagnosis nor are there any standardized therapeutic strategies. Diagnosis of ME/CFS relies on subjective clinical evaluation of patient-reported symptoms and exclusion of other chronic conditions that could explain the symptoms (2, 6).

Since the COVID-19 pandemic, millions of individuals have been diagnosed with Long COVID (LC) (7, 8). Given the remarkable similarity in symptoms shared by some individuals with LC and ME/CFS, there is speculation that the illnesses share disease mechanisms (9). The rise in LC has sparked an urgent global need to better understand these conditions together and independently. To maintain the distinction between LC and ME/CFS that is not initiated by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), this study focuses solely on patients diagnosed with ME/CFS prior to the onset of the COVID-19 pandemic.

Several recent lines of evidence have implicated immune system dysregulation in ME/CFS pathogenesis and/or pathophysiology (10, 11). Peripheral immune cells, including both innate and adaptive populations, exhibit abnormalities in ME/CFS (12). Single-cell RNA sequencing (scRNA-seq) of peripheral blood mononuclear cells (PBMCs) from ME/CFS cases and sedentary controls has revealed significant dysregulation in classical monocytes and T cells, with markers of T cell exhaustion and chronic inflammation observed in patients (13, 14). Other studies have reported alterations in monocyte and natural killer (NK) cell surface markers (15, 16), increased neutrophil apoptosis (17), and metabolic and cytokine production abnormalities in T lymphocytes (18). Dysregulation of B cells has also been implicated (19, 20). Furthermore, plasma levels of pro- and anti-inflammatory cytokines, including elevated transforming growth factor beta (21), correlate with ME/CFS disease severity and duration (22–24). Despite these

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emerging insights, current methods for diagnosing ME/CFS rely on symptomology due to the lack of diagnostic molecular markers (25).

Circulating cell-free RNA (cfRNA) is a promising bioanalyte to explore ME/CFS pathobiology and diagnostic strategies. cfRNA is released into the bloodstream during cellular turnover and reflects dynamic changes in gene expression, intercellular signaling, and tissue-specific processes (26, 27). Plasma-derived cfRNA provides a minimally invasive window into cellular health and has shown potential to predict, diagnose, and monitor diverse medical conditions, as well as distinguish between causes of inflammatory disease (28, 29). Here, we examine plasma cfRNA for ME/CFS cases and healthy sedentary controls at baseline, independent from symptom provocation through exercise. By profiling cfRNA, this research aims to uncover disease-specific biomarkers, characterize immune system dysfunction, and advance our understanding of ME/CFS pathobiology to inform diagnoses and treatment strategies.

Results

ME/CFS Cohort for cfRNA Sequencing. The ME/CFS case–control cohort analyzed in this study was part of an initiative led by the Cornell-NIH ME/CFS Center (30). We collected a total of 175 plasma samples from three test sites: Ithaca College, Weill Cornell Medicine in New York City, and the ID Med clinic in Los Angeles (Fig. 1 *A* and *C*). Patients with reported comorbidities or samples that did not meet the cfRNA quality control criteria were excluded (*Materials and Methods*). A total of 168 samples were included in the analysis (n = 93 confirmed ME/CFS cases, n = 75 sedentary controls, *SI Appendix*, Fig. S1).

To capture the heterogeneity of symptoms seen in ME/CFS patients, the cohort included cases with a range of disease severities, assessed using self-reported SF-36 general health (GH) and multidimensional fatigue inventory (MFI-20) scores (*Materials and Methods* and Dataset S1). GH survey scores ranged from 0 (worst health) to 100 (perfect health), whereas MFI-20 ranged from 100 (highest fatigue) to 0 (no fatigue). GH was significantly lower in cases for both male and female subjects (Fig. 1*B*). As expected, GH and MFI-20 were inversely correlated, with higher fatigue scores associated with lower GH scores. Disease duration among cases varied widely, spanning 6 mo to 38 y (Fig. 1*B*). A control cohort included sedentary individuals matched for sex, age, and VO_2 peak (Fig. 1*D*). Lastly, the cohorts were predominantly female (approximately 75%; n = 126 females, n = 42 males) reflecting the greater prevalence of females within the ME/CFS population.

We isolated and sequenced cfRNA from plasma to generate circulating cfRNA profiles for each subject. We obtained an average of 40.8 million reads per sample (range: 6.1 to 79.1 million). The mean number of gene features, or reads assigned to annotated genes, per sample was 9.0 million (range: 1.2 to 41.4 million), with a mean alignment rate of 50.7% (range: 13 to 83%). The alignment rates are consistent with those reported in previous cfRNA studies across various disease contexts (29, 31, 32). While these rates are lower than what is typically observed in whole-blood transcriptomics, this is expected given the lower RNA biomass in

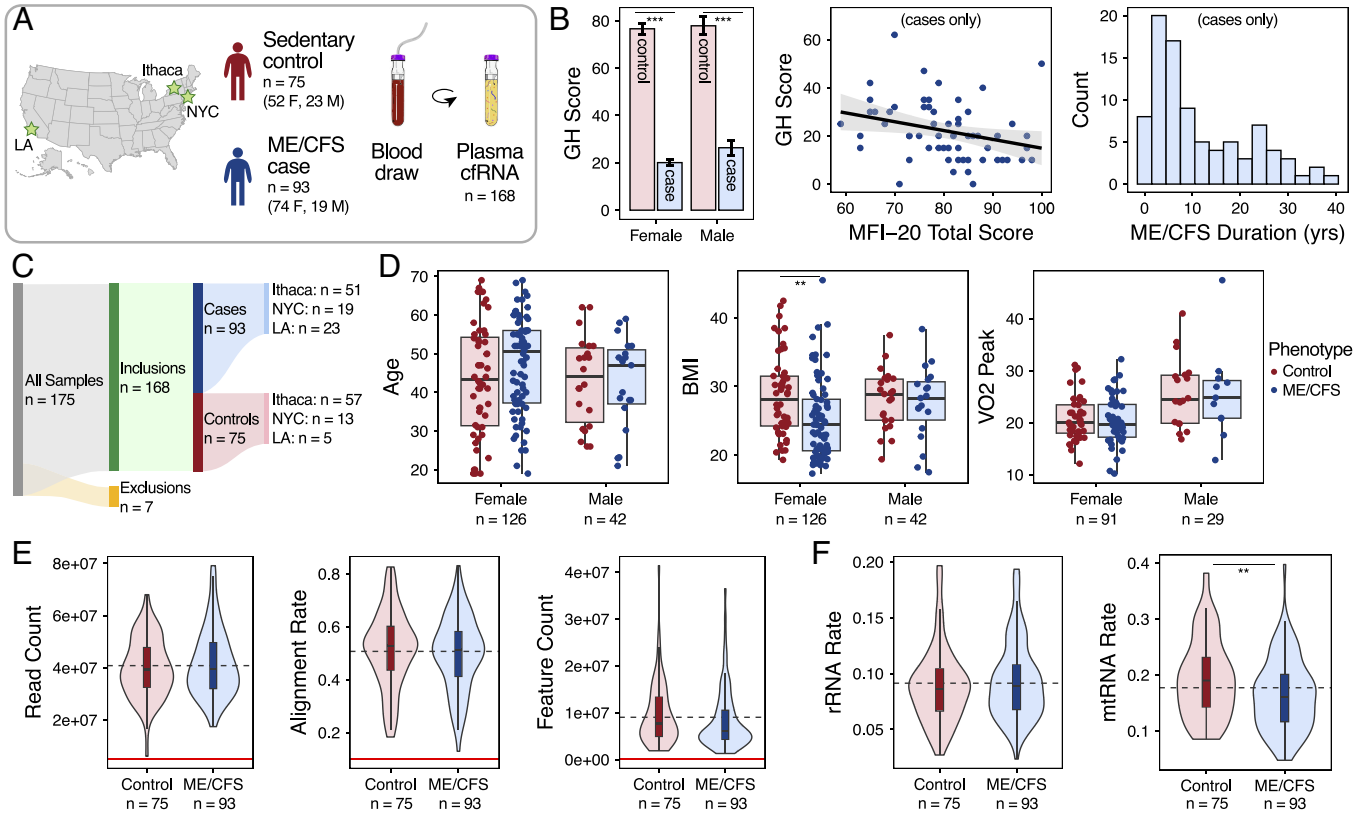


Fig. 1. Cohort metrics of controls and cases. (*A*) Case and control subjects were recruited to Los Angeles (n = 30), Ithaca (n = 110), and New York City (n = 35) to collect ~500 µL plasma samples for cfRNA analysis (n = 168 passed QC, *Materials and Methods*). The map illustration was created using BioRender (https://biorender.com). (*B*) Clinical parameters of cohort. General health (GH) score is significantly higher in controls. GH score and multidimensional fatigue inventory (MFI-20 Total) for cases are inversely correlated (Pearson r = −0.3, *P*-value < 0.03). Histogram plot of disease duration in cases (range 6 mo to 38 y). (*C*) Composition of cohort subjects. (*D*) Comparison of age, BMI, and VO_2 peak by sex and phenotype. VO_2 peak measurements obtained via cardiopulmonary exercise testing (CPET) as part of a larger study (n = 120). (*E*) Comparison of cfRNA sequencing total read counts, alignment rate, and total feature counts, by phenotype. (*F*) Comparison of rRNA and mtRNA rates before filtering between phenotypes. Dashed lines indicate mean value of all samples. Asterisks indicate statistical significance by Mann–Whitney *U* test, unless otherwise noted, using *P*-values as follows; **P* < 0.05, ***P* < 0.01, ****P* < 0.001, or not significant if left blank.

plasma and reflects the unique biological nature of cfRNA (27). Sequencing output metrics, including total reads, feature counts, and alignment rates, did not differ significantly between cases and controls (Fig. 1*E*). We also observed a mean ribosomal RNA (rRNA) rate of 9.1% and a mean mitochondrial RNA (mtRNA) rate of 17.7% across all samples. While the rRNA rate did not differ significantly between phenotypes, the mtRNA rate was significantly lower in cases (mean of 19.4% in controls and 16.4% in cases; Mann–Whitney *U* test *P*-value < 0.01, Fig. 1*F*).

cfRNA Biomarker Discovery for an ME/CFS Diagnostic Tool. To date, no clinically validated biomarkers for objective diagnosis of ME/CFS have been established. Current diagnostic methods rely primarily on clinical symptom assessment, and occasionally, cardiopulmonary exercise testing (CPET) (2, 33). Liquid biopsies analyzing circulating cfRNA present an opportunity for unbiased and minimally invasive diagnostic approaches. We implemented differential abundance analysis (DESeq2, *Materials and Methods*) and identified 743 unique features that differed between cases and controls [Benjamini–Hochberg (BH) adjusted *P*-values (*P*-adj) threshold of < 0.05 and a $\log_2$ fold change cutoff of ±0.5]. Of these, 608 features were elevated in cases, and 135 were elevated in controls. Genes such as IL18R1, CCR7, FCRL5, PRF1, and IFNLR1 were more abundant in cases, whereas genes such as CCL3, TGFB2, CD109, and COL1A1 were more abundant in controls (Fig. 2*A*). In addition, we observed changes in abundance of several mtRNA transcripts between cases and controls. Notably, levels of a mitochondrial ribosome assembly factor (MTG1) were significantly increased, while several transcripts (MTND1P23, MT-TL1, MT-TV, MT-TH, MT-TS2, MT-TT, MTATP8P1, MTND6P3, and MTRNR2L3) were significantly decreased. Many of these mtRNA transcripts are inversely correlated with MTG1 in cases (*SI Appendix,* Fig. S2). These findings support a role for mitochondrial dysfunction in ME/CFS, with dysregulated mtRNA expression potentially contributing to altered energy metabolism (34–36).

Unsupervised clustering based on the differentially abundant features demonstrated separation between cases and controls (Fig. 2*B*). To build machine learning classifiers, we used Monte Carlo cross-validation by repeatedly partitioning the dataset into training (70%) and test (30%) sets, while balancing for phenotype, sex, and test site. This process was repeated 100 times, generating 100 unique train–test splits (*Materials and Methods*). From each training set, we selected features using differential abundance criteria (*P*-value < 0.001 and base mean > 50) and trained 15 machine learning models (Fig. 2*C*). This approach yielded approximately 1,500 models. We evaluated performance based on test set accuracy, sensitivity, specificity, and area under the receiver operating characteristic curve (AUC-ROC), assessing both the average metrics and the best-performing seed for each model (Fig. 2*D*).

As expected, tree-based algorithms such as ExtraTrees, Random Forest (RF), and C5 exhibited strong training performance, reflecting their capacity to fit complex patterns. However, these models tended to overfit the training data as evidenced by poor performance on the test set. Models with robust performance demonstrated high accuracy and AUC-ROC values for both the training and test sets. Across all models, each sample was included in the test set approximately 450 times (~30% of all iterations). We observed variability in individual classification rates, with some samples classified correctly >90% of the time, while others classified correctly as low as 11% of the time (Fig. 2*E*). This result suggests that certain samples possessed unique features which drove consistent correct or incorrect classification.

We highlight one particularly promising model identified based on the highest test AUC-ROC. This generalized linear model with regularized regression using the least absolute shrinkage and selection operator (GLMNET Lasso) achieved a training AUC-ROC of 0.86 (95% CI: [0.789 to 0.932]) and a testing AUC-ROC of 0.81 (95% CI: [0.694 to 0.926], Fig. 2*F*), with case–control separation determined by a Youden threshold of 0.65. The training set reached 80% accuracy, 71% sensitivity, and 92% specificity, whereas the test set reached 77% accuracy, 84% sensitivity, and 68% specificity (Fig. 2*G*). This GLMNET Lasso identified a 21-gene signature, including genes with roles in RNA processing, transcriptional regulation, immune response, and metabolic regulation—key processes implicated in ME/CFS pathophysiology. HNRNPLL, a regulator of alternative splicing and mRNA stability in immune pathways, was the strongest contributor to model performance (Fig. 2*H*). The model performance was consistent across sample characteristics, with no significant differences observed with test site or sex (Fig. 2*I*). Notably, severity metrics, including GH scores for all samples and MFI-20 scores for cases, were not correlated with classification performance (Fig. 2*J*).

To confirm that models were trained on ME/CFS-specific features, we conducted a phenotype permutation test in which we randomly shuffled the sample phenotype labels (Fig. 2*K*). As expected, most models exhibited a sharp drop in AUC-ROC following label shuffling. This effect was particularly pronounced for the selected GLMNET Lasso classifier, which exhibited a substantial performance decline in both training and test sets, reinforcing the biological relevance of the identified cfRNA signatures (Fig. 2*L*). The only exception was the RPART (recursive partitioning and regression trees) model which showed improved test performance on shuffled data, a clear indication of overfitting.

Bioinformatic Deconvolution of cfRNA Cell Types of Origin. Vu et al. (2024) recently used scRNA-seq of PBMCs to demonstrate immune cell dysregulation in ME/CFS patients (14). To examine this through the lens of plasma cfRNA, we performed deconvolution of the cell types that contribute cfRNA to the plasma mixture. We used a Bayesian algorithm for deconvolution (BayesPrism, *Materials and Methods*) (37) and the ME/CFS PBMC scRNA-seq atlas created by Vu et al. as a reference (Fig. 3*A*) (14). This atlas spanned 28 cell types from 58 case–control subjects, 40 of which overlapped with our cohort (Fig. 3*B*). Although this reference lacked solid tissue cell types, when we performed deconvolution with the Tabula Sapiens single-cell reference (38), the contribution of solid tissue-derived cfRNA was minimal and no significant differences were observed between phenotypes (*SI Appendix,* Fig. S3).

Using this approach, we identified six cell types that differed significantly between cases and controls (*P*-adj < 0.05, Fig. 3*C*). In order of significance, these included plasmacytoid dendritic cells (pDCs), monocytes (Fig. 3*B*; cluster 26), naïve CD8+ T cells, T cells (Fig. 3*B*; cluster 16), mucosal-associated invariant T (MAIT) cells, and platelets (Fig. 3*D*). To understand deconvolution differences by sex, we first analyzed only female subjects (n = 126) and identified six cell types with significant differential abundance (Fig. 3*E*). While statistical comparisons between female and male ME/CFS cases remained challenging due to sample size, we identified some cell types with patterns that suggested potential sex-based differences in immune responses. For instance, we found effector/memory (E/M) CD8+ T cells (Fig. 3*B*; cluster 5) to be significantly elevated only in female cases versus controls and not significantly different when we included both sexes. Additionally, we saw classical monocytes as elevated in both female cases and controls compared to their male counterparts (Fig. 3*E*).

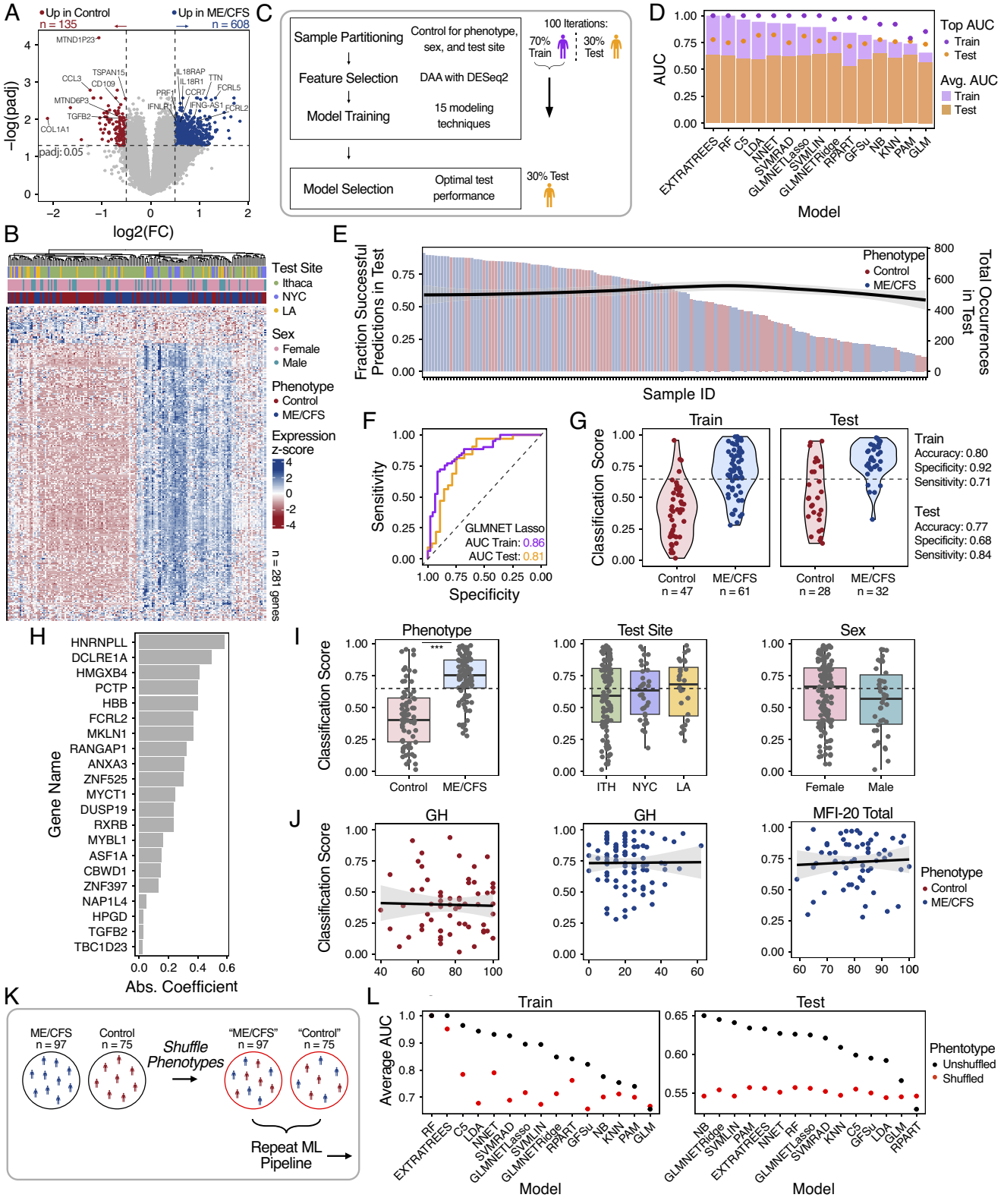


Fig. 2. Machine learning for ME/CFS classification. (*A*) Volcano plot of differential abundance analysis of exercise-independent cfRNA between cases and controls (DESeq2). 608 transcripts are elevated in cases (blue, BH adjusted *P*-values (*P*-adj) < 0.05, $\log_2$FoldChange > 0.5) and 135 transcripts are elevated in controls (red, *P*-adj < 0.05, $\log_2$FoldChange < −0.5). (*B*) Variance stabilizing transformation (VST) values of significantly differentially abundant genes (n = 281 genes, *P*-adj < 0.05, abs($\log_2$FoldChange) > 0.75) between phenotypes. Samples and genes are clustered based on correlation. (*C*) Workflow of machine learning pipeline. (*D*) Bar chart of average performance across 100 iterations for each machine learning model for train (purple) and test (orange) sets. Dots indicated the top-performing iteration for each model. (*E*) Fraction of successful classification for each subject in the test set (left *y*-axis). Number of times each sample is assigned to the test set (right *y*-axis) shown by a LOESS smoothed line (black) across samples color coded by phenotype. (*F*) AUC-ROC plot of model performance for one GLMNET Lasso train (purple) and test (orange) set distinguishing cases from controls. (*G*) Violin plots of classifier scores for training (specificity: 0.92, sensitivity: 0.71, accuracy: 0.80) and test sets (specificity: 0.68, sensitivity: 0.84, accuracy: 0.77). Youden threshold cutoff (0.65) shown with dashed line. (*H*) 21 gene features selected by the GLMNET Lasso ranked by absolute coefficient. (*I*) Comparison of model performance and phenotype (***$P < 1e-12$), test site, and sex. Dashed lines indicate Youden threshold (y = 0.65). (*J*) Correlation of model performance and disease severity determined by MFI-20 and GH for cases (blue) or controls (red). (*K*) Workflow for phenotype permutation test. (*L*) Average AUC across each model before (black) and after (red) shuffling the sample phenotypes.

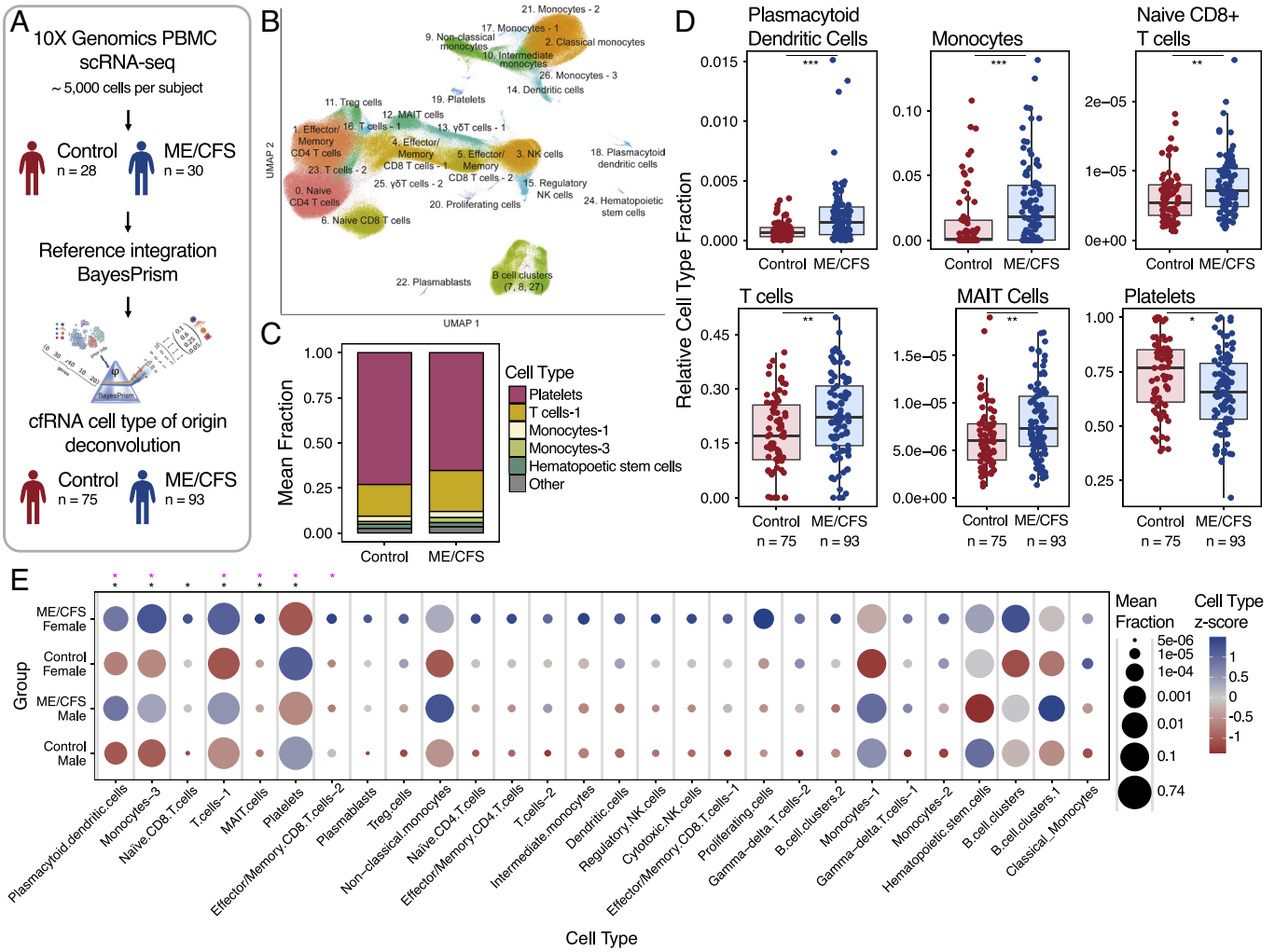


Fig. 3. Bioinformatic cell type of origin deconvolution of plasma cfRNA. (*A*) Deconvolution workflow using a scRNA-seq PBMC reference from the ME/CFS cohort and BayesPrism (12). (*B*) Uniform manifold approximation and projection (UMAP) from Vu et al. (14) depicts 28 PBMC cell type clusters defined by Seurat (11). (*C*) Average predicted fraction of cfRNA derived from the top five cell types and others between phenotypes. (*D*) Topmost significantly different cell type proportions between phenotypes (*P*-adj values as follows; $*P < 0.05$, $**P < 0.023$, $***P < 0.004$). (*E*) Dot plot comparing the relative mean cell type fraction (dot size) and the cell type z-score (dot color) within each cell type for cases and controls separated by sex (* indicates significance $P < 0.05$, black: both sexes, pink: only females, or not significant if left blank).

Biological Networks Underlying ME/CFS. To gain insight into the biological systems underlying ME/CFS pathobiology, we performed Ingenuity Pathway Analysis (IPA, *Materials and Methods*). The top canonical pathways enriched in cases versus controls, ranked by –log(*P*-value) and |*z*-score|, included focal adhesion kinase signaling, NFKBIE signaling, lipid antigen presentation by CD1, IL-4 signaling, neural cell adhesion molecule signaling, and T cell exhaustion signaling (Fig. 4*A*). These pathways are closely linked to immune dysregulation, chronic inflammation, altered cell signaling, and impaired energy metabolism, all hallmarks of ME/CFS biology (13, 39–43). Interestingly, the pathway we identified with the most decreased activity in cases compared to controls was CTLA4 signaling in cytotoxic T lymphocytes. This is a critical pathway involved in immune checkpoint activity and prevention of overactivation of the immune system (40). This finding suggests potential differences in immune tolerance or suppression mechanisms in ME/CFS cases compared to controls. Network analysis of the significantly different pathways revealed key biological themes in ME/CFS, including: 1. Cytokine signaling driven by CSF2, IFNG, IL2, IL21, and TNF. 2. Immune cell activation and stimulation with an increase in leukocytes, T cells, and other immune cells. 3. JAK-STAT pathway involving STAT1 and STAT4 as immune regulators (Fig. 4*B*). Collectively, this network reflects widespread immune dysregulation in ME/CFS.

To further explore these pathways, we generated a heatmap using 69 marker genes from the top 10 pathways activated in ME/CFS as well as the most inhibited pathway (Fig. 4*C*). Additionally, we analyzed the cfRNA abundance levels of the 20 most significantly different pathway related genes between cases and controls (Fig. 4*D*). Among these, three genes, ADAM12, TGFB2, and MMP2, were significantly reduced in ME/CFS cases compared to controls, while the remaining 17 genes, including, IL18R1, TNFSF15, ITGA10, and several T cell receptor genes, were elevated in cases. These changes in transcript abundance suggest an altered immune response and potential dysregulation of tissue repair and inflammation pathways in ME/CFS.

Absence of Consistent Viral cfRNA Signatures Associated with ME/CFS. To investigate the potential presence of circulating viral RNA, we performed cfRNA metagenomic analysis (Kraken2, Bracken, *Materials and Methods*). We first mapped all cfRNA reads that aligned to the human genome to a database of microbial references. After filtering reads from known contaminants (*Materials and Methods* and *SI Appendix*, Fig. S4), the top three most

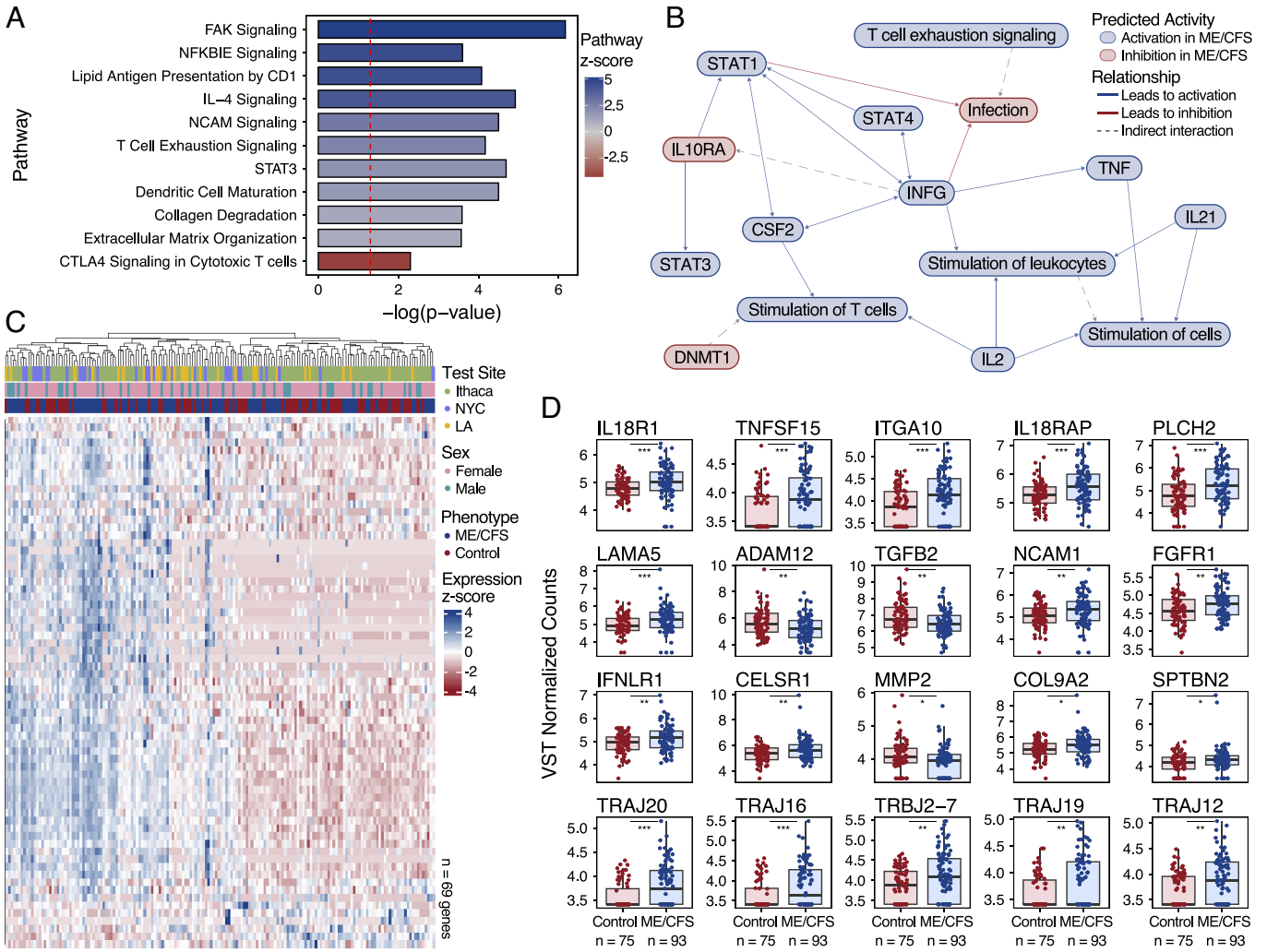


Fig. 4. Biological pathway analysis of ME/CFS. (*A*) Top 10 most significantly activated (blue) pathways and most significantly inhibited (red) pathways in cases as determined by −log(*P*-value) and a nonzero z-score. (*B*) Network analysis of the predicted pathways of ME/CFS (Qiagen, IPA). (*C*) Heatmap of activated (blue) and inhibited (red) pathway marker genes for the selected pathways (n = 69 genes). (*D*) VST counts of the top 20 most significantly differentially pathway marker genes between controls and cases (*P*-adj values as follows; **P* < 0.05, ***P* < 0.01, ****P* < 0.005).

abundant viral families detected were *Retroviridae*, *Herpesviridae*, and *Flaviviridae* (Fig. 5*A*). However, we found no significant differences in the $\log_{10}$ transformed counts per million (CPM) +1 values of total viral burden between phenotypes (Fig. 5*B*) nor did we observe differences in the burden of individual viral families (Fig. 5*C*). Additionally, there was no significantly different detectable signal from the *Picornaviridae* family, which includes enteroviruses (Fig. 5*D*), a viral genus that has been suspected in the onset of ME/CFS (44–46).

Interestingly, we identified one case and one control who exhibited highly elevated levels of total viral reads (>4 $\log_{10}$(CPM +1), Fig. 5*B*). These reads mapped primarily to the *Flaviviridae* family, specifically the *Pegivirus hominis* (HPgV) virus (Fig. 5*C*). HPgV infections are common and typically asymptomatic. HPgV has been shown to modulate immune responses and interact with other viral infections such as HIV but has not been associated with a specific disease (47). The cfRNA reads derived from HPgV accounted for 10.5% of the total reads in the case subject, and

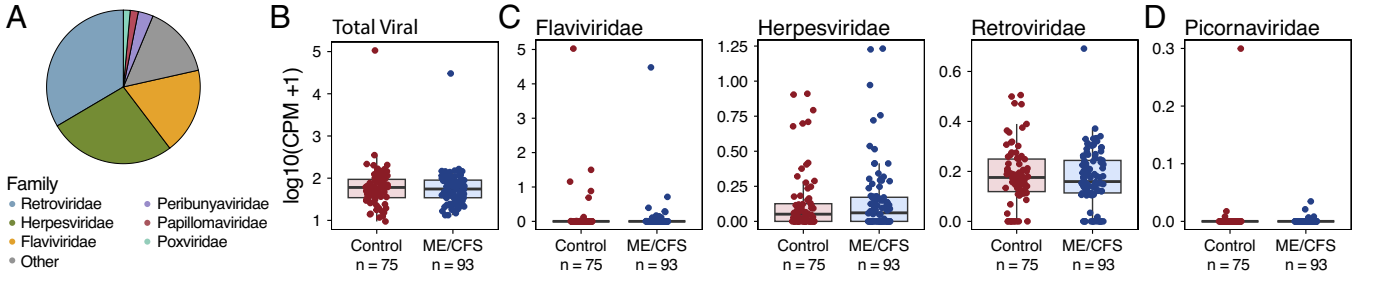


Fig. 5. Detection of circulating viral cfRNA. (*A*) Top six and other (gray) most abundant viral families with background viral reads removed (Kraken2, Bracken). (*B*) Comparison between phenotypes of $\log_{10}$ scaled CPM +1 sum of all viral reads (total viral) after background removal. (*C*) $\log_{10}$ scaled (CPM) +1 sum of top three individual viral families (*Flaviviridae*, *Herpesviridae*, and *Retroviridae*) between phenotypes were not significantly different. (*D*) $\log_{10}$ scaled (CPM) +1 sum of Enterovirus family (*Picornaviridae*) was not significantly different.

3.0% of the total reads in the control subject (Fig. 5*C*). Apart from these individual samples, the overall viral burden detected by cfRNA in cases and controls was low, and no consistent viral signatures were detected within the cohort.

Discussion

There is a growing need for objective, minimally invasive, and accurate tools for diagnosing chronic illnesses. Developing a laboratory test for ME/CFS has been particularly challenging due to the complex range of patient-reported experiences (48). Currently, the primary diagnostic strategy for ME/CFS relies on meeting clinical symptom criteria and excluding other similarly presenting illnesses.

Advances in omics measurements have enabled researchers to identify potential biomarkers for ME/CFS (30, 48–52). However, due to small sample sizes and disease heterogeneity, no study has generated a definitive diagnostic tool. In this study, we investigated the potential of RNA liquid biopsy as a tool for both diagnosis and characterization of the pathobiology of ME/CFS. Recent research has demonstrated that cfRNA can predict, diagnose, and monitor other complex inflammatory and infectious diseases. Some of these include pediatric conditions, such as Kawasaki disease and multisystem inflammatory syndrome in children (32), complications following hematopoietic stem cell transplantation (53, 54), COVID-19 (28), and tuberculosis (29).

The use of an untargeted platform for plasma-derived RNA sequencing has not previously been reported in ME/CFS. Using unbiased cfRNA sequencing, we identified significant differences in the abundance of over 700 transcripts in cases versus controls. We developed a GLMNET Lasso model with a train and test AUC of 0.86 and 0.81, respectively. This model selected 21 cfRNA features, including TGFB2, FCRL2, and HNRNPLL, which are related to cytokine response, B cell function, and mRNA stability, particularly in T cell activation. Other selected features are broadly involved in transcriptional regulation, cellular stress, and oxygen transport. Pathway analysis highlighted key biological processes associated with ME/CFS, including dysregulated cytokine signaling, immune cell activation, and the JAK-STAT signaling pathway. Enrichment of pathways such as T cell exhaustion, IL-4 signaling, and lipid antigen presentation aligns with impaired immune regulation and chronic inflammation. Additionally, the downregulation of the CTLA4 signaling pathway suggests possible deficits in immune checkpoint control, which may contribute to prolonged inflammation.

We observed that across all samples in the test sets, there was an uneven distribution of successful classification predictions. Certain samples, when placed into the test set, were classified correctly in more than 90% of the models, while others were classified correctly in as few as 11%. We found this distribution to be dependent not on a clinical metric, such as phenotype, nor test site, but rather on the fraction of platelet-derived cfRNA (*SI Appendix*, Fig. S5). The relative platelet fraction of each sample was determined by cell type of origin deconvolution. We further showed that this correlation was not due to a difference in sample collection protocols across set sites, but a true biological difference in platelets between cases and controls. This finding supports the notion that ME/CFS patients experience platelet dysfunction at baseline (14).

Our findings support the hypothesis that immune dysregulation underlies ME/CFS pathology. Through deconvolution analysis with a disease-specific scRNA-seq reference from Vu et al. (14), we identified significant differences in cfRNA contributions from multiple immune cell types. Notably, we observed a decrease in platelet-derived cfRNA and an increase in pDCs, monocytes, and certain T cell subset-derived cfRNA in cases compared to controls. Previous studies, including scRNA-seq analyses with our matched cohort, also show similar trends of improper platelet activation in ME/CFS patients (14, 55). The most significantly elevated cfRNA cell type of origin in cases versus controls are pDCs. pDCs are a critical component of the innate immune system and are responsible for producing type I interferons in response to viral infections. Activity of pDCs has been implicated in various autoimmune and inflammatory diseases, and their activity here suggests a heightened interferon-mediated antiviral response and persistent immune stimulation in ME/CFS patients. Similarly, monocytes displayed increased cfRNA fraction in cases. Monocytes are key mediators of inflammation and tissue repair, and their heightened presence may indicate sustained immune activation and aberrant monocyte trafficking. Prior studies have also reported altered monocyte transcriptomes and inflammatory cytokine production in ME/CFS (13, 14, 24, 50).

Among T cell subtypes, we found significant differences in naive CD8+ T cells, MAIT cells, and other T cell clusters. Naive CD8+ T cells are essential for adaptive immunity, and their increased cfRNA contribution in cases, although unexpected as these are an unactivated and relatively stable cell type, may suggest altered T cell homeostasis or impaired differentiation into effector and memory subsets. MAIT cells, which comprise a T cell subset involved in mucosal immunity and microbial defense, have been increasingly recognized for their role in chronic inflammatory diseases (11). We also observed an increased contribution of E/M CD8+ T cells in females with ME/CFS, suggesting potential sex-specific immune alterations. E/M CD8+ T cells play a crucial role in long-term immune surveillance and rapid responses to antigen re-exposure, and their elevation may indicate impaired resolution of inflammation and/or chronic immune activation in female cases.

Our cfRNA metagenomic analysis did not detect significant differences in viral RNA burden between phenotypes. Although viral reads mapping to *Retroviridae*, *Herpesviridae*, *Flaviviridae*, and others were detected, their levels were not significantly altered between phenotypes. Nevertheless, given that few of our reads were from solid tissue, we cannot rule out the presence of viral reservoirs whose cfRNA was not captured from the circulation. Most reports of persistent virus in ME/CFS involve tissues and organs rather than blood (46).

This study has limitations. Most importantly, our analyses focused solely on baseline samples to examine exercise-independent differences between phenotypes. Future studies could extend this work by investigating cfRNA dynamics during PEM in ME/CFS patients, in contrast to healthy individuals who do not experience adverse effects after exercise. Such work would have the potential to provide insight into the biological responses that lead to PEM as cfRNA reflects both immune and tissue injury through cell turnover.

In summary, our results provide evidence of immune dysregulation in ME/CFS and highlight key pathways that may contribute to pathobiology. Given the observed immune shifts and altered mtRNA profiles, potential therapeutic avenues for ME/CFS may include targeting inflammatory pathways, type I interferon signaling, or mitochondrial function. Expanding the cohort size will also be crucial for capturing patient heterogeneity, ultimately improving the robustness and generalizability of cfRNA-based biomarkers for ME/CFS.

Materials and Methods

Patient Cohort. The human subject research described here is part of a multisite initiative of the Cornell Collaborative Research Center for the study of ME/CFS. All human subject protocols were approved by the Institutional Review Boards

of Weill Cornell Medical College (Protocol 1708018518) and Ithaca College (IRB 1017-12Dx2). Written informed consent was obtained prior to screening and written informed consent was obtained prior to medical review and the subsequent protocols as previously described (30). Participants accepted into the study completed testing at one of three sites: Ithaca College, Weill Cornell Medicine in New York City, and the ID Med clinic in Los Angeles. Subjects provided demographic information, donated blood, and many participated in CPETs as part of the larger study (33). Each patient fulfilled the Institute of Medicine and Canadian Consensus Criteria for diagnosis (6) and was required to have disease onset prior to the emergence of the SARS-CoV-2 pandemic. This cohort was selected using sex, age, body mass index (BMI), and VO_2 peak from CPETs performed on the day of sample collection (33, 56). Controls were screened for sedentary behavior, which was quantified using the VO_2 peak measurement, a measure of cardiovascular fitness, as previously described (30) (Fig. 1*D*). Sex was matched to the approximate disease prevalence being greater in females than males (57, 58). Chi-squared tests were used to evaluate the difference between sex within the cases and controls. Age ranges were matched between and within the phenotype groups and BMI was matched for the males only, as determined by Mann–Whitney *U* Test (Fig. 1*D*).

Several surveys were employed to measure disease severity in this cohort, regarding fatigue (59), GH (60), and PEM (61). These surveys have been described previously (14).

Plasma Collection. Whole blood was collected from each participant in K2 ethylenediaminetetraacetic acid (EDTA) tubes (n = 175 samples) and left at room temperature for no more than 4 h before processing. Plasma was separated by centrifugation at 500×g for 5 min at room temperature and immediately stored in 1 mL aliquots at −80 °C. Transfer of plasma occurred on dry ice, and samples were stored at −80 °C until getting processed.

Plasma cfRNA Isolation and Library Preparation. Prior to cfRNA library preparation, plasma was thawed at room temperature, then ~500 μL samples were aliquoted and centrifuged at 3,000× g for 8 min at 4 °C. cfRNA libraries were prepared for sequencing using methods previously described (29), with the following modification: Each pool was sequenced using the Illumina NovaSeq 6000 platform (paired-end, 150 bp).

Bioinformatic Processing and Filtering. Sequencing results were processed using a custom bioinformatics pipeline as previously described (29), with the following modifications: Reads were not trimmed to 61 bp prior to processing and samples were filtered based on total read counts, alignment rate, total feature counts, DNA contamination, and RNA degradation (62–65) (*SI Appendix,* Fig. S1*A*). Samples which passed the quality filter all had greater than five million reads, greater than 100,000 feature counts, greater than 10% alignment rate, less than onefold intron to exon ratio, less than three SD from the mean of the 5′ to 3′ bias, and less than 95% duplication rate. We confirmed that sample collection date had minimal impact on cfRNA sequencing quality metrics, including total read count, alignment rate, and 5′ to 3′ bias. While collection date was statistically associated with 5′ to 3′ bias, the correlation was weak (r = 0.17) (*SI Appendix,* Fig. S1*B*).

Differential Abundance Analysis. Comparative analysis of differentially abundant genes was performed similarly to methods previously reported (28), with the following modifications: a BH adjusted *P*-value (*P*-adj) threshold of <0.05 and a $\log_2$ fold change cutoff of ±0.5 was applied (66). Biological pathways were identified using QIAGEN IPA software (v01-23-01).

Machine Learning. Machine learning and model training were performed using DESeq2 (v1.34.0) (66), Caret33 (v6.0.90) (67), and pROC34 (v1.18.0) (68) R packages. The sample metadata and count matrices were first split into a training set (70%) and a test set (30%), which were evenly partitioned across phenotype, sex, and test site. We repeated the split 100 times, for 100 different iterations of the train and test groups. Within each of the 100 training sets, feature selection was performed using differential abundance analysis which selected genes with a base mean of greater than 50 and a *P*-value less than 0.001. These genes were used to train 15 machine learning algorithms using fivefold cross-validation and grid search hyperparameter tuning. For each iteration of each model, accuracy, sensitivity, and AUC-ROC were used to measure test performance.

The 15 classification models include generalized linear models with Ridge and LASSO feature selection (GLMNET Ridge and GLMNET Lasso), support vector machines with linear and radial basis function kernel (SVMLin and SVMRAD), RF, RF ExtraTrees (EXTRATREES), neural networks (NNET), linear discriminant analysis (LDA), nearest shrunken centroids (PAM), C5.0 (C5), k-nearest neighbors (KNN), naive bayes (NB), CART (RPART), generalized linear models (GLM), and very greedy forward search algorithm (GFSu), as described previously (29).

Cell Type Deconvolution. Cell type of origin deconvolution of cfRNA was performed using BayesPrism (37) (V3.1.3) with a ME/CFS PBMC single-cell RNA-seq atlas as a reference (14). As described previously, approximately 5,000 PBMCs per sample, from 30 case and 28 control subjects, were analyzed using the 10× Genomics platform. Of the 58 subjects, 40 are also included in our cfRNA study cohort. The cell types were grouped into 29 clusters using Seurat (v4.1.0). The maximum count of each cluster was input into BayesPrism as the reference.

Deconvolution was also performed using the Tabula Sapiens single-cell RNA-seq atlas (Release 1) as a reference. Cells from this atlas were grouped as previously described in Vorperian et al. (69). Cell types with more than 100,000 unique molecular identifiers were included in the reference and subsampled to 300 cells using ScanPy (v1.8.1) (70).

Metagenomic Classification. Metagenomic classification of unaligned reads was performed using Kraken2 with the default database and parameters. Bracken was used to quantify species level abundances, and a custom python script was used for formatting. To normalize the count outputs from Bracken, we divided the new estimated counts by the number of sequenced reads and multiplied that by one million for CPM. Count values were $\log_{10}+1$ transformed. The following background viral families (nonmammalian infecting) were removed: Siphoviridae, Baculoviridae, Tospoviridae, Autographiviridae, Alphaflexiviridae, Virgaviridae, Tymoviridae, Bromoviridae, Myoviridae, Solemoviridae, Genomoviridae, Endornaviridae, Tombusviridae, Casjensviridae, Herelleviridae, Chrysoviridae, Phycodnaviridae, Straboviridae, Potyviridae, Marnaviridae.

Quantification and Statistical Analysis. All statistical analyses were performed using R (v4.2.3). All statistical analyses were performed using R (v4.1.0). Data wrangling and visualization were performed using Python (3.9.1), Pandas (1.3.0) matplotlib-venn (0.11.6), R (v4.1.0), Tidyverse (v1.3.1), and ggplot2 (v3.3.5). Bioinformatic pipelines were run using the Snakemake workflow management system (71). Statistical significance was tested using Mann–Whitney *U* tests, unless otherwise stated. All sequencing data were aligned to the GRCh38 Gencode v38 Primary Assembly and features counted using the GRCh38 Gencode v38 Primary Assembly Annotation (62). Visualizations were created using Adobe Illustrator 2024 software (v.28.0).

Data, Materials, and Software Availability. All codes have been deposited to GitHub (72). Deidentified RNA sequencing data has been deposited in the NIH/National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO) repository through GEO Series accession number GSE293840 (73).

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